Searching for the Fake Weight

Grade 8: Weight and Capacity

Introduction

In this activity, we can use the understanding of a weighing scale to search for a particular weight. This can be done in many ways, and some of these methods will be faster than others. This is similar to how a computer also searches. In this activity, we will explore search algorithms using a weighing balance and see how we can speed up the process using different approaches. We will separate a lighter fake weight from 8 proper weights among 9 identical-looking weights. We convey to students that the same reasoning can be applied to the analysis of computer algorithms.

Activity	Time	Description
Launch	5 min	The teacher poses the question to the students and invites a couple of students to try their ideas. Activity Video Link: https://tinyurl.com/searchingactivityvideo
Problem-solving and algorithm	25 mins	Student's Tryout the activity with each other. Student Worksheet: https://tinyurl.com/searchingactivityworksheet As the teacher progresses through the activities, the students progress along with the worksheet.
Discussions and Explorations	5 min	The discussion on the various algorithms, how a slight change in approach leads to a faster solution, and how computers apply such algorithms

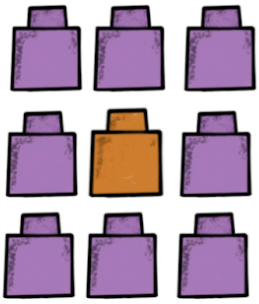
CT Components

This activity develops in a way that helps students discover important ideas, such as breaking a problem into smaller problems (decomposition), identifying patterns, developing algorithms, and finding faster ways to complete a task, thus analyzing algorithms. These are fundamental ideas involved in computational thinking.

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Activity: Searching with a Weighing Scale

Imagine we have **9 identical-looking weights** out, of which one is **Lighter, a fake weight**. We have to figure out which one is lighter using only a weighing scale (and nothing else).
Our goal is to find the fake in the least number of weighings.





How will we figure this out? What process will we use to determine the fewest weighings needed to identify the lighter fake weight? Always assume that you have the worst luck and will get the fake weight in the last possible step.

Make the students observe these things:

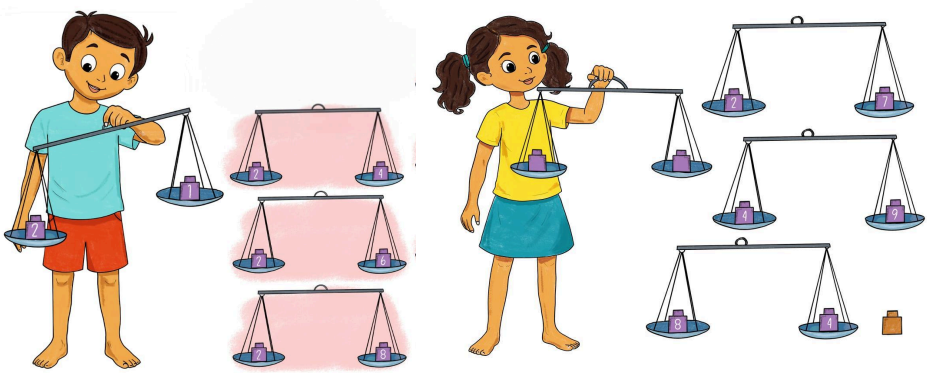
- A 2-pan weighing balance can be in 3 positions during weighing. Either the left or the right side could be heavier, or both sides could be balanced.
- We weigh two weights, one on either side, and the weighing balance shows that they are equal. From this, we can conclude that. The fake weight is in the remaining weights.
  We know that there is only one fake weight that is lighter than the rest. This means all the other weights weigh the same. So, if the weighing scale is balanced, the weights are not fake. This implies that the lighter weight is in the rest of them.
- By observing how many weights can be discarded after each weighing, we can determine which method is faster. The method that helps us remove more weight per round is faster than the rest of the methods

1: Initial attempt
Let's take **two weights** at a time and compare. Taking two weights at a time can be done in two different ways:

- In the first method, we keep the weight on **one side constant** (let's say the right) and choose **a new weight** for the left side at each weighing.
- Second method, we chose **two new weights** for every weighing.

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Demonstrate both methods partially to the students. Do not completely show the students the methods. Or you can invite a couple of students whose answers seem like a good method to come and do it in front of the class.



1. Is one of the methods above faster (fewer weighings) than the other, and if yes which one?
    a. Yes, method one is faster
    b. Yes, method two is faster
    c. No, both take the same number of weighings

Correct answer: c. Method two is faster
Explanation: In method one, we can remove only one weight per weighing, whereas in the second method, we can remove two weights. Removing more weights in each weighing means fewer weights to compare in future rounds.
Competency: Analysis, observations, deduction

Demonstrate both methods to the students so that they can verify their answers.

2. How many minimum weighings will it take to guarantee figuring out the fake (lighter) weight when you compare two weights at a time?
   (This means we are assuming you always have bad luck and get the fake one in the last possible step)
    a. 5 weighings
    b. 8 weighings
    c. 4 weighings
    d. 7 weighs

Correct answer: 4 weighings
Explanation: We compare two weights; if they are equal, we know neither is fake, so we remove both. Similarly, we have to compare two more pairs. Now, in the fourth attempt, we have 3 weights remaining. If the two chosen weights are unequal, then the lighter one is the one we have found; if they are equal, the remaining weight is the lighter one. This type of strategy is called Divide and Conquer.
Competency: Pattern recognition, logic

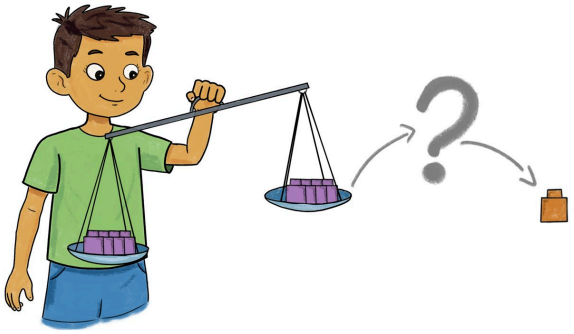
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For the first method suggested above, you have to compare one fixed weight with the remaining 8 weights to find the one that differs in weight. Using this method, we can figure out the lighter weight in 7 weighings (Why not 8?). This is a Linear search.

Key observations after the above activity:

- If weights are equal when weighed, then they are both not the fake weight (Why?)
- We can remove a maximum of 2 weights in each round
- We need to remove the maximum possible weights in each round, and divide-and-conquer is more efficient than linear/sequential search.
- If we have 3 weights remaining, and compare any two. Then, if the two chosen weights are unequal, then the lighter one is the fake; if they are equal, the remaining weight is the fake weight.

2: A faster way (Binary search)



To do it faster, we need to remove as many weights as possible in each round and use divide-and-conquer better. We can divide the 9 weights into 2 equal parts of 4 each, leaving 1 weight.
In the first weighing, we keep 4 on the left and 4 on the right, and the remaining is set aside.

3. After one weighing, how many weights remain?
    a. 1
    b. 4
    c. 5
    d. 8

Correct answer: 4 weights remain in consideration

Explanation: When we compare 4 weights with 4 other weights, the weighing scale is either balanced or not balanced. In the worst-case situation, they are not balanced. This means the lighter side will be above the other 4. So we can remove the 4 weights on the heavier side, and we can also remove the one we didn't weigh, since it's not a fake weight either. 5 proper weights get removed from the search space.

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Competency: Algorithmic thinking, decomposition, generalization

4. If the fake is among the 4 remaining weights, how many more weighings are needed (minimum) to find the fake weight?
    a. 1
    b. 2
    c. 3
    d. 4

Correct answer: 2 more weighings.
Explanation: We divide the 4 weights into 2 groups of 2 and compare them. One side will be lighter than the other; it will have the fake weight. So we have 2 weights left. We further split these two weights, compare them, and then find the fake weight.

Competency: algorithmic thinking, analysis, decomposition

We figured out the fake weight in 3 weighings. This method of dividing a problem into two parts and then finding a solution is known as Binary search.
Can we find the fake weight in fewer weighings than this?

3: Fastest possible algorithm in a three-state system

Can we do this faster than the binary search? Let's bring together the various points we observed above. We saw that the weighing balance is a three-state system. We also saw that we can infer information about a weight without actually weighing it. We also saw that with three weights, we can infer which one is the fake in a single weighing.

Let's try dividing into 3 equal parts, and now we will weigh two groups of three.



5. How many weights get removed after the first weighing?
    a. 2

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b. 4
c. 3
d. 6

Correct answer: 6 weights.
Explanation: When we compare the two groups of 3 weights, they will be either balanced or not balanced. If the weighing scale is balanced, then the 6 weights on the balance are not fake. Otherwise, one of the sides with the three weights is lighter, which implies that the 3 on the heavier side and the three not on the balance are not fake. Either way, we remove 6 weights.

Competency: algorithmic thinking, analysis, decomposition, pattern recognition

6. How many steps does this method require to determine the fake weight?
    a. 1
    b. 2
    c. 3
    d. 4

Correct answer: 2 weighings.
Explanation: In the first weighing, we removed two-thirds of the 9 weights; 3 weights remain. Similarly, we remove 2 of the 3 weights in the second weighing and thus identify the fake weight in 2 weighings (refer to the observations made after the first step and the explanation for the previous question).

Competency: algorithmic thinking, analysis, decomposition, observation, pattern recognition

We see that if we have three equal groups, we can find the lightest group in one step. If these groups each have three weights, then in the second weighing, we get the lightest weight.

Conclusion
We see that dividing the search space into three parts is faster for finding the fake weight using a balance. This happens because in a three-state system, “ternary search” is a faster searching algorithm.

We broke the problem into 3-weight chunks, understood the pattern, saw how it gets solved, and used the same algorithm to solve the bigger 9-weight problem.

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